Cardiac Biomarkers

AMI markers: ideal properties, kinetics, interpretation and case patterns

LAB401 taken biomarker timeline

Definition and ideal biomarker

- A measurable blood substance indicating heart muscle damage or stress.
- Ideal: high cardiac specificity, high sensitivity, quick rise, long diagnostic window, clinical correlation with infarct size/prognosis, accurate rapid cost-effective assay.
- No single marker is perfect, so multi-marker approaches were used.

How biomarkers appear in blood

- Necrotic myocytes lose membrane integrity.
- Intracellular macromolecules diffuse into microvasculature and lymphatics.
- These macromolecules are detected in blood.
- Current MI definition requires clinical myocardial ischemia plus rise/fall in cardiac troponin values.

AMI diagnosis shift

- 1979 diagnosis emphasized typical ischemic chest pain >20 minutes, ECG changes, and rise/fall in CK/CK-MB.
- 2018 definition: myocardial ischemia plus cTn rise/fall, with evidence such as thrombus on angiography/autopsy or imaging evidence of new viable myocardium loss/new regional wall motion abnormality.

Marker Kinetics

Marker	Initial rise	Peak	Back to normal
Total CK	4-8 hr	12-24 hr	3-5 days
CK-MB	3-8 hr	12-24 hr	2-3 days
LDH	10-12 hr	48-72 hr	5-14 days
AST	6-12 hr	18-36 hr	4-6 days
Myoglobin	1-3 hr	6-9 hr	1 day
cTnI	3-8 hr	24-48 hr	3-5 days
cTnT	3-8 hr	24-48 hr and 72-100 hr	5-10 days

General enzyme-testing guidance

- Draw blood promptly after symptom onset.
- Use serial determinations at appropriate intervals.
- Choose enzyme tests based on the time gap since infarct onset.
- ECG remains key when presentation is very early and biomarkers are still normal.

Enzyme biomarkers

- CK: found in skeletal muscle, myocardium, brain; rises when damaged myocytes leak CK. Total CK can rise from skeletal muscle injury.
- CK isoenzymes: CK-MM skeletal, CK-MB cardiac, CK-BB brain. CK-MB >6% total CK plus total CK >2x suggests MI.
- CK-MB remains useful for re-infarction, infarct size estimation, and where troponin assay is unavailable.
- LDH is widespread and non-specific; LD-1 can help late diagnosis.
- AST is non-specific because it is in liver, heart, skeletal muscle, kidney, RBCs.

Protein biomarkers

- Myoglobin: cardiac and skeletal muscle protein; very early and sensitive, but poor specificity. Useful negative predictor and reperfusion monitoring.
- Troponins: TnC calcium-binding, TnI inhibitory, TnT tropomyosin-binding.
- Cardiac-specific troponins have high sensitivity, early rise similar to CK-MB, and prolonged elevation useful for late presenters.

Case Interpretation Patterns

Case	Pattern	Meaning
Very early chest pain + STEMI	At 1 hr CK-MB, myoglobin, troponin may still be normal.	Does not rule out AMI; ECG is key; repeat biomarkers at 3 and 6 hr.
Late presentation, symptoms 5 days ago	CK-MB and myoglobin normal; troponin I elevated.	Likely MI days earlier because troponin persists longer.
Chest pain after weightlifting	CK-MB mildly high, myoglobin markedly high, troponin normal.	Most likely skeletal muscle injury/rhabdomyolysis, not cardiac injury.